

Value Added Problems

Problem 1: Print all values in the BST within a given range

Problem 1: BST Range Query

```
#include <stdio.h>

#include <stdlib.h>

struct Node {

    int data;

    struct Node* left;

    struct Node* right;

};

struct Node* createNode(int val) {

    struct Node* node = (struct Node*)malloc(sizeof(struct Node));

    node->data = val;

    node->left = node->right = NULL;

    return node;

}

void rangeQuery(struct Node* root, int low, int high) {

    if (root == NULL) return;

    if (root->data > low)

        rangeQuery(root->left, low, high);
```

```
if (root->data >= low && root->data <= high)

    printf("%d ", root->data);

if (root->data < high)

    rangeQuery(root->right, low, high);

}
```

```
int main() {

    printf("Name: Sharannya Gholse\n");

    printf("PRN: 25070521118\n");

    printf("Batch: B2\n\n");

    struct Node* root = createNode(10);

    root->left = createNode(5);

    root->right = createNode(15);

    root->left->left = createNode(3);

    root->left->right = createNode(7);

    root->right->right = createNode(18);

    int low = 7, high = 15;

    printf("Nodes in range: ");

    rangeQuery(root, low, high);

    printf("\n");

}
```

```
    return 0;
}
```

Problem 2: Check if expression is balanced

Problem 2: Balanced Brackets (C)

```
#include <stdio.h>
#include <string.h>

#define MAX 1000000

char stack[MAX];
int top = -1;

void push(char c) {
    stack[++top] = c;
}

char pop() {
    return stack[top--];
}

int isEmpty() {
    return top == -1;
```

```
}
```

```
int isBalanced(char* s) {  
    for (int i = 0; s[i] != '\0'; i++) {  
        char c = s[i];  
  
        if (c == '(' || c == '{' || c == '[')  
            push(c);  
        else {  
            if (isEmpty()) return 0;  
  
            char t = pop();  
  
            if ((c == ')' && t != '(') ||  
                (c == '}' && t != '{') ||  
                (c == ']' && t != '['))  
                return 0;  
        }  
    }  
}
```

```
    return isEmpty();  
}
```

```
int main() {  
    char s[] = "{[()]}"  
    printf("Name: Sharannya Gholve\n");  
    printf("PRN: 25070521118\n");  
    printf("Batch: B2\n\n");  
  
    if (isBalanced(s))  
        printf("Balanced\n");  
    else  
        printf("Not Balanced\n");  
  
    return 0;  
}
```